26-BS-PH101 — Physics

Course Code	26-BS-PH101	**Category**	Basic Science Course
Course Title	Physics	**L – T – P**	3 – 0 – 0
Credits	3	**Contact Hours**	36
Semester	First or Second (by group)	**Assessment**	Theory: CA 30 + ESE 70

Prerequisites

Builds on the Higher Secondary Physics of WBCHSE, CBSE or ISC. Under the group rotation it is taken in one semester by one group and in the other by the second; the syllabus is identical. Assumed:

- Oscillations, waves and the elements of optics (CBSE XI Units VII, X; WBCHSE; ISC).
- Electrostatics, current electricity and magnetism (CBSE XII Units I–IV).
- The elements of modern physics — the photoelectric effect and the Bohr atom (CBSE XII Units VIII–IX).
- Calculus to the level of Mathematics-I, taken concurrently.

Course Objectives

- To establish the physics of oscillations and waves as the basis of much of engineering.
- To develop the electromagnetic theory and the wave picture of light used across the branches.
- To introduce quantum and solid-state physics as the foundation of modern devices.
- To connect each principle to an engineering application and to laboratory measurement.

Course Outcomes

On completion of the course the student will be able to:

CO	Statement	Bloom
CO1	Apply the theory of simple harmonic, damped and forced oscillations and of waves to physical systems.	L3 – Apply
CO2	Apply the principles of interference, diffraction and polarisation to optical phenomena and instruments.	L3 – Apply
CO3	Apply the laws of electromagnetism, including Maxwell's equations in outline, to fields and waves.	L3 – Apply
CO4	Apply the elements of quantum and solid-state physics to the behaviour of materials and devices.	L4 – Analyse
CO5	Relate the theory to measurement, estimating the uncertainty of a physical quantity determined in the laboratory.	L4 – Analyse

CO–PO Articulation Matrix

Correlation: 3 substantial, 2 moderate, 1 slight, – none. Mapped against the eleven Programme Outcomes of the NBA (GAPC v4.0).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11
CO1	3	2	1	1	1	–	–	–	–	–	1
CO2	3	2	1	2	1	–	–	–	–	–	1
CO3	3	3	2	2	2	1	1	–	–	–	1
CO4	3	2	2	2	2	1	1	–	–	–	2
CO5	3	2	1	3	2	–	–	1	1	–	2

Course Plan — Module-wise Syllabus

Module	Title	Contents	Hours
I	Oscillations and Waves	Simple harmonic motion from the differential equation; damped and forced oscillations, resonance and the quality factor; coupled oscillators in outline; the wave equation, travelling and standing waves; superposition, beats and the Doppler effect.	8
II	Wave Optics	Interference — Young's experiment, thin films, Newton's rings; diffraction — single slit, the grating and its resolving power; polarisation — Malus' law, double refraction, the quarter- and half-wave plate; the elements of the laser and of optical fibre.	8
III	Electromagnetism	Electrostatics and magnetostatics in review; the laws of Gauss, Faraday and Ampère; the displacement current; Maxwell's equations in integral and differential form (statement and meaning); the electromagnetic wave in vacuum in outline.	8
IV	Quantum Physics	The failure of classical physics; the photoelectric and Compton effects; the de Broglie hypothesis and wave–particle duality; the uncertainty principle; the Schrödinger equation and the particle in a box; the quantum numbers in outline.	7
V	Solid-State and Modern Physics	Free-electron theory of metals in outline; the band theory of solids and the distinction of conductor, semiconductor and insulator; intrinsic and extrinsic semiconductors; superconductivity in brief; an introduction to nanomaterials and their engineering use.	5

Cross-cutting notes. The physics laboratory (BS-PH191/291) is taken concurrently and is assessed separately; the theory is to be sequenced so that the relevant principle precedes the experiment. Module V connects forward to the Basic Electrical and Electronics course and to the materials content of the disciplines.

Text and Reference Books

- A. B. Bhattacharya, Introduction to Electromagnetic Theory (with Lab Manual), Khanna Publishing House (AICTE Prescribed Textbook)
- A. Beiser, Concepts of Modern Physics, McGraw-Hill.
- D. J. Griffiths, Introduction to Electrodynamics, Pearson.
- Halliday, Resnick and Walker, Fundamentals of Physics, Wiley.
- A. B. Bhattacharya, Engineering Physics, Khanna Publishing House
- B. B. Laud, Electromagnetics, New Age.
- A. Ghatak, Optics, McGraw-Hill.

Suggested SWAYAM / NPTEL and Online Resources

- NPTEL — Engineering Physics I / II (IITs), 12 weeks.
- NPTEL — Waves and Oscillations, 8 / 12 weeks.
- NPTEL — Semiconductor Physics and Devices, 12 weeks.
- Virtual Labs (Ministry of Education) — Optics, Modern Physics and Electricity & Magnetism laboratories.

Question-paper note. At least one question in Group C is to be a numerical problem — a resonance, interference-fringe, field or quantum calculation — with units and significant figures required. Standard constants are printed with the paper.